

Universal AI: Solomonoff Induction

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- **The common engine**: a single **Bayesian mixture** ξ over a class $\mathcal{M}$ of candidate environments, weighted by a prior w_ν .
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- **Universal** choice: $\mathcal{M}$ = all computable environments, prior $w_\nu = 2^{-K(\nu)}$ (K is the “complexity” of the environment). Assumption “ $\mu \in \mathcal{M}$ ” becomes “*the universe is computable*”.

Part I: Solomonoff Induction

Learning to predict

The problem of induction, formalized

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- **Quality measure:** Cumulative μ -expected squared prediction error $S_\infty^\mu := \sum_{t=1}^\infty s_t^\mu$

Per-step squared prediction error s_t^μ

$$s_t^\mu = \mathbb{E}_\mu \left[\sum_{x_t} (P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}))^2 \right] = \sum_{x_{<t}} \mu(x_{<t}) \left[\sum_{x_t} (P(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}))^2 \right]$$

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- **Goal:** build a P that makes cumulative error S_∞^μ *small*.
- **Idea:** don't bet on one model. Hedge your bets and mix over a whole class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ of potential environments.

The Bayesian mixture ξ

Definition

Class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ with prior weights $w_\nu > 0$, $\sum_\nu w_\nu = 1$.

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- **Posterior update is multiplicative:** $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \frac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$: we reweight by how well ν predicted vs. ξ (Bayes' rule!)

Main result: the cumulative prediction bound

Cumulative bound

$$S_{\infty}^{\mu} := \sum_{t=1}^{\infty} \sum_{x_{<t}} \mu(x_{<t}) \sum_{x_t \in \mathbb{B}} (P(x_t | x_{<t}) - \mu(x_t | x_{<t}))^2 \leq -\ln w_{\mu}$$

- Total squared error over *all time* is finite, bounded by the prior penalty on μ .
- Higher prior on the truth $\Rightarrow$ smaller total error.

Specializing to the Solomonoff prior

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Explicit complexity bound

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- **Interpretation:** the total prediction error is bounded by the *description length* of the true environment. Simpler worlds are learned faster.
- This is the formal version of **Occam's razor**.

Other properties of the Solomonoff prior

- **Bounded mistakes:** on a deterministic sequence, predicting the most likely bit $\hat{x}_t = \arg \max_{x \in \mathbb{B}} \xi(x \mid x_{<t})$ makes at most $-2 \ln w_\mu$ wrong guesses *ever*.

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- **Pareto optimality:** no other predictor beats ξ on *every* $\nu \in \mathcal{M}$ simultaneously (for KL loss *and* squared loss).
- **Misspecification** ($\mu \notin \mathcal{M}$): the bound degrades gracefully,

$$S_\infty^\mu \leq -\ln w_{\hat{\mu}} + D_n(\mu \parallel \hat{\mu})$$

where $\hat{\mu}$ is the closest model in $\mathcal{M}$ to μ .

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- **Why care for safety:** a precise, assumption-light model of idealized intelligence providing a lens on what optimal agents do. For example, we can show under what circumstances AIXI would [wirehead](#) [RO11], [safely rather than recklessly explore](#) [CCH20], or [self-modify](#) [OR11] as a model of what superintelligent agents would do.

References I

- [CCH20] Michael K. Cohen, Elliot Catt, and Marcus Hutter.
Curiosity killed or incapacitated the cat and the asymptotically optimal agent.
CoRR, abs/2006.03357, 2020.
URL: <https://arxiv.org/abs/2006.03357>.
- [OR11] Laurent Orseau and Mark Ring.
Self-modification and mortality in artificial agents.
In *Artificial General Intelligence (AGI 2011)*, volume 6830 of *Lecture Notes in Computer Science*, pages 1–10.
Springer, 2011.
URL: https://link.springer.com/chapter/10.1007/978-3-642-22887-2_1, doi:10.1007/978-3-642-22887-2_1.
- [RO11] Mark Ring and Laurent Orseau.
Delusion, survival, and intelligent agents.
In *Artificial General Intelligence (AGI 2011)*, volume 6830 of *Lecture Notes in Computer Science*, pages 11–20.
Springer, 2011.
URL: <https://people.idsia.ch/~ring/AGI-2011/Paper-B.pdf>, doi:10.1007/978-3-642-22887-2_2.